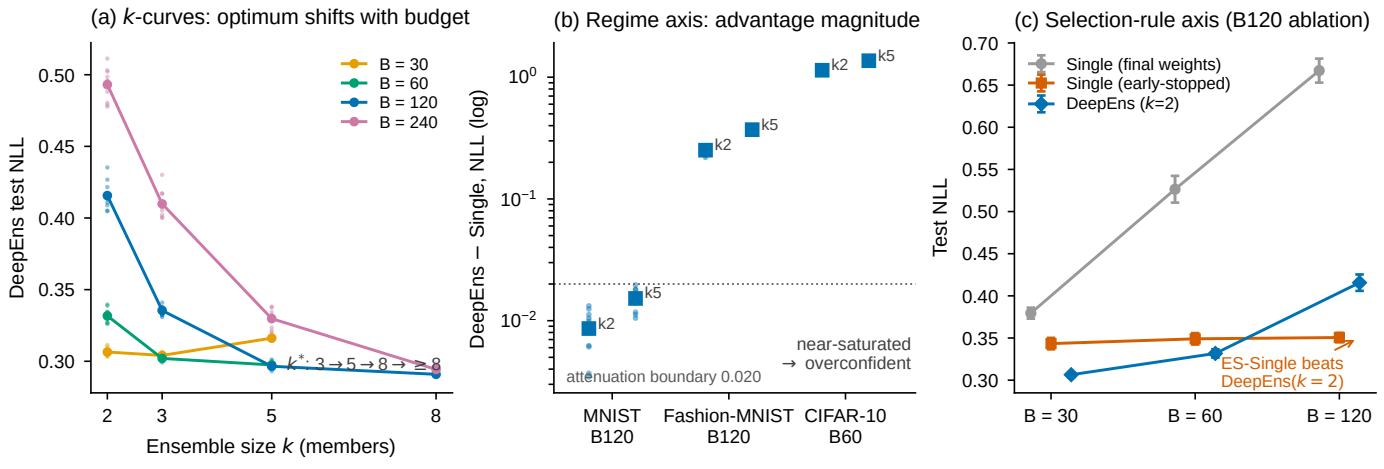
(a) k-curves: optimum shifts with budget
DeepEns test NLL
0.50
0.45
0.40
0.35
0.30
B = 30
B = 60
B = 120
B = 240
k*: 3 → 5 → 8 → ≥ 8
2
3
5
8
Ensemble size k (members)
(b) Regime axis: advantage magnitude
DeepEns − Single, NLL (log)
10^0
10^−1
10^−2
k2
k5
k2
k5
k2
k5
near-saturated
→ overconfident
attenuation boundary 0.020
MNIST
B120
Fashion-MNIST
B120
CIFAR-10
B60
(c) Selection-rule axis (B120 ablation)
Test NLL
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
Single (final weights)
Single (early-stopped)
DeepEns (k=2)
ES-Single beats
DeepEns(k = 2)
B = 30
B = 60
B = 120